

Node Structure Visualization

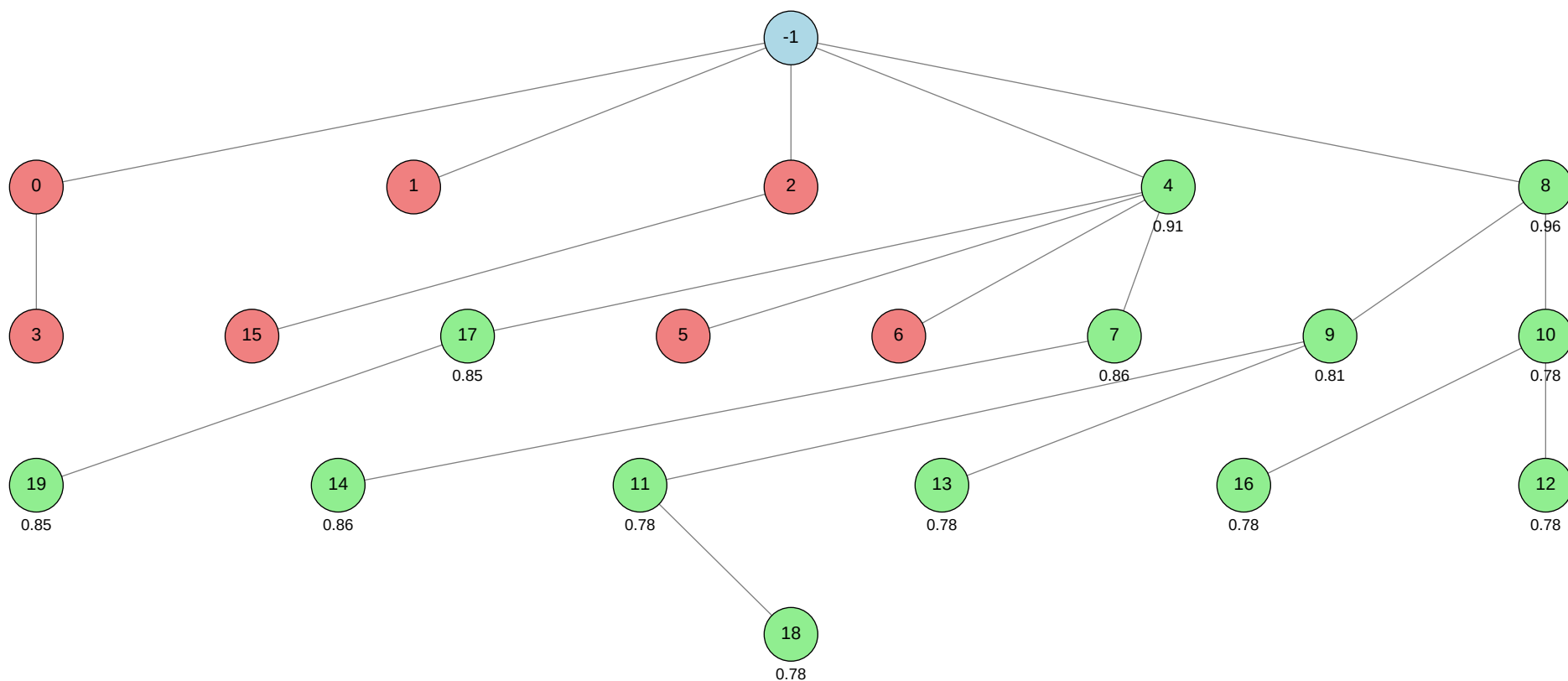
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9606 (Node 8)

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
<-1			
>0			autogluon.multimodal
>1			machine learning
>2			autogluon.tabular
>3			autogluon.multimodal
>4			machine learning
>5			machine learning
>6			machine learning
>7			machine learning
>8			autogluon.multimodal
>9			autogluon.multimodal
>10			autogluon.multimodal
>11			autogluon.multimodal
>12			autogluon.multimodal
>13			autogluon.multimodal
>14			machine learning
>15			autogluon.tabular
>16			autogluon.multimodal
>17			machine learning
>18			autogluon.multimodal
>19			machine learning



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	0.0520
ctime	2026-05-22 16:55:19
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	7
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	5
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	10.786520000000001
validated_visits	13
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 4, 8]

Node 0 (evolve)

Property	Value
uct_value	1.7306
ctime	2026-05-22 16:56:16
debug_attempts	1
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.4418091240986957
failure_penalty	-0.0
failure_visits	2
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	2
parent_id	-1
child_ids	[3]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 33-basic_prompt/node_0/conda_env

Resolved 235 packages in 2.37s

Downloading pydantic-core (2.0MiB)

Downloaded pydantic-core

Prepared 10 packages in 2.54s

Uninstalled 1 package in 1.83s

Installed 233 packages in 1m 11s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + ... [truncated, 8149 chars total]

Error Analysis:

ERROR_SUMMARY: The script failed with `ValueError: Unknown label encountered: benign` because the label mapping function (`map\_label\_to\_binary`) only recognizes "malignant" and "non-neoplastic" as valid labels. However, the training data contains "benign" as a label, which is not handled by the mapping function, causing the error when it tries to process a "benign" label.

SUGGESTED_FIX:

Update the `map\_label\_to\_binary` function to recognize "benign" as a valid non-malignant class by mapping both "benign" and "non-neoplastic" to 0. Review the training data for any other possible label variants and ensure all non-malignant labels are mapped to 0, while "malignant" is mapped to 1, to prevent similar errors.

Node 1 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-22 17:04:16
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2486433288776608
failure_penalty	-0.0
failure_visits	1
id	1
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

warning: Requirements file `~/home/anri21/.conda/envs/mlzero\_env/lib/python3.11/site-packages/autogluon/assistant/tools\_registry/machine learning/requirements.txt` does not contain any dependencies

Using Python 3.11.15 environment at: 33-basic_prompt/node_1/conda_env

Resolved 79 packages in 585ms

Uninstalled 1 package in 1.65s

Installed 78 packages in 47.34s

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.1

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ filelock==3.29.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

+ huggingface-hub==1.16.1

+ idna==3.16

+ jinja2==3.1.6

```
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+... [truncated, 6926 chars total]
```

Error Analysis:

ERROR_SUMMARY: The main error is that the training process exceeded the allowed wall-clock time limit of 3600 seconds (1 hour), causing the script to be forcibly terminated before model training completed (only ~52% of the final epoch was finished). The repeated OSError: [Errno 16] Device or resource busy messages are related to attempts to delete temporary files on an NFS-mounted filesystem and are cleanup warnings, not the root cause of the failure.

SUGGESTED_FIX:

Reduce the total runtime by decreasing the number of training epochs (e.g., from 5 to 2 or 3), lowering the batch size, or using a smaller model architecture (e.g., resnet18 instead of a larger model). Additionally, consider simplifying data augmentations or using fewer workers in the DataLoader to reduce overhead. Monitor training progress and ensure that the script completes all steps (training, inference, and saving outputs) well within the 1-hour time limit.

Node 2 (evolve)

Property	Value
uct_value	1.7306
ctime	2026-05-22 18:07:36
debug_attempts	1
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	0.8578388191546724
failure_penalty	-0.0
failure_visits	2
id	2
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	2
parent_id	-1
child_ids	[15]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 33-basic_prompt/node_2/conda_env

Resolved 239 packages in 3.06s

Uninstalled 1 package in 1.79s

Installed 237 packages in 43.58s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2

+ hyperopt==... [truncated, 42641 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon failed to train any models because Ray (a parallelization backend used by AutoGluon) attempted to create UNIX domain sockets with file paths exceeding the system limit of 107 bytes. This resulted in repeated OSError exceptions: "validate_socket_filename failed: AF_UNIX path length cannot exceed 107 bytes", causing all model training to be skipped and ultimately raising RuntimeError: "No models were trained successfully during fit()".

SUGGESTED_FIX:

Shorten the base directory paths used for temporary files and model outputs to ensure all UNIX socket paths remain under 107 bytes. Move your working directory (including data, scratch, and output folders) to a location with a much shorter absolute path (e.g., /tmp/ml or /scratch/ml), and re-run the script. If using a cluster or shared filesystem, coordinate with your administrator to ensure temporary directories (like \$TMPDIR or /tmp) are used for Ray's socket files.

Node 3 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 18:14:27
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	3
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	3
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	0
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 33-basic_prompt/node_3/conda_env

Resolved 235 packages in 1.16s

Uninstalled 1 package in 1.98s

Installed 233 packages in 49.12s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.1

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.1... [truncated, 15232 chars total]

Error Analysis:

ERROR_SUMMARY: The root cause is a KeyError: "None of [Index(['Unnamed: 0'], dtype='object')] are in the [columns]", which occurs when the code tries to select the 'Unnamed: 0' column from the test DataFrame after it has already been dropped. This happens because the code assumes the presence of identifier columns (like

'Unnamed: 0') in the test set, but after dropping them, it still attempts to access them, leading to the error. The OSError: Device or resource busy errors are related to NFS file system cleanup and are non-fatal, not affecting the main logic.

SUGGESTED_FIX:

Update the logic that determines the identifier columns (``id_cols``) for the test set to ensure it only includes columns that actually exist after dropping index columns like 'Unnamed: 0' and 'index'. Before selecting columns for the output DataFrame, check that each column in ``id_cols`` is present in the DataFrame; if not, skip or remove it from the list. This will prevent KeyErrors when constructing the output.

Node 4 (debug)

Property	Value
uct_value	1.2608
avg_raw_score	0.8693424999999999
ctime	2026-05-22 18:39:55
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.3284111976269926
exploration	0.7429102097403913
failure_penalty	-0.0
failure_visits	2
id	4
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.49261679644048895
num_children	4
prev_tutorial_prompt	None
stage	debug
time_step	4
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.3284111976269926
validated_reward	4.32649

validated_visits	5
validated_weight	0.6666666666666666
validation_score	0.9120
visits	7
parent_id	-1
child_ids	[17, 5, 6, 7]

Node 5 (evolve)

Property	Value
uct_value	1.9725
ctime	2026-05-22 19:24:41
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	evolve
time_step	5
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 33-basic_prompt/node_5/conda_env
Resolved 79 packages in 591ms
Uninstalled 1 package in 1.89s
Installed 78 packages in 44.45s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.16.1
- + idna==3.16
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.14.0a1
- + pydantic-core==2.47.0
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+... [truncated, 8754 chars total]

Error Analysis:

ERROR_SUMMARY: The main error is that the script was terminated after exceeding the 1-hour (3600 seconds) wall-clock time limit, so model training, prediction, or output validation did not complete. The repeated "OSError: [Errno 16] Device or resource busy: '.nfs...'" messages are secondary and occur during cleanup, likely due to open file handles on an NFS-mounted filesystem.

SUGGESTED_FIX:

Reduce the total runtime by decreasing the model complexity (e.g., use a smaller architecture or fewer epochs), lowering batch size, or using fewer data augmentations. Additionally, ensure that all file handles are properly closed before script exit to minimize NFS cleanup errors, but focus primarily on shortening training and inference time to stay within the 1-hour limit.

Node 6 (evolve)

Property	Value
uct_value	1.9725
ctime	2026-05-22 20:27:33
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.7938516450278406
failure_penalty	-0.0
failure_visits	1
id	6
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have then
stage	evolve
time_step	6
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 33-basic_prompt/node_6/conda_env
Resolved 79 packages in 372ms
Uninstalled 1 package in 1.85s
Installed 78 packages in 44.05s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.16.1
- + idna==3.16
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.14.0a1
- + pydantic-core==2.47.0
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+... [truncated, 9007 chars total]

Error Analysis:

ERROR_SUMMARY: The primary error is that the process was terminated after exceeding the 1-hour (3600 seconds) wall-clock time limit, causing model training to be interrupted before completion and preventing predictions or validation from being performed or saved. The repeated OSError: [Errno 16] Device or resource busy messages are secondary, likely resulting from attempts to clean up temporary files that were still in use due to the abrupt termination.

SUGGESTED_FIX:

Reduce the total runtime by decreasing the model complexity, lowering the number of training epochs (e.g., from 4 to 1–2), reducing batch size or image resolution, or using a smaller model architecture. Additionally, ensure that data loading and augmentation are efficient (e.g., lower num_workers if I/O is a bottleneck), and consider saving intermediate results or checkpoints to avoid total loss if the time limit is reached.

Node 7 (debug)

Property	Value
uct_value	1.8315
avg_raw_score	0.859295
ctime	2026-05-22 21:30:23
debug_attempts	1
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.4367352614015575
exploration	1.3383640602871656
failure_penalty	-0.0
failure_visits	0
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.4367352614015575
num_children	1
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug

time_step	7
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.4367352614015575
validated_reward	1.71859
validated_visits	2
validated_weight	1.0
validation_score	0.8605
visits	2
parent_id	4
child_ids	[14]

Node 8 (debug)

Property	Value
uct_value	1.0140
avg_raw_score	0.8075037500000002
ctime	2026-05-22 22:12:13
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.14868604004449507
exploration	0.8578388191546724
failure_penalty	-0.0
failure_visits	0
id	8
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.14868604004449507
num_children	2
prev_tutorial_prompt	None
stage	debug
time_step	8
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.14868604004449507
validated_reward	6.4600300000000015

validated_visits	8
validated_weight	1.0
validation_score	0.9606
visits	8
parent_id	-1
child_ids	[9, 10]

Node 9 (evolve)

Property	Value
uct_value	1.0669
avg_raw_score	0.7921266666666668
ctime	2026-05-22 22:36:43
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.06316277345198436
exploration	1.138807119808647
failure_penalty	-0.0
failure_visits	0
id	9
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.06316277345198436
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	9
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.06316277345198436
validated_reward	3.15715
validated_visits	4
validated_weight	1.0
validation_score	0.8148
visits	4
parent_id	8
child_ids	[11, 13]

Node 10 (evolve)

Property	Value
uct_value	1.1772
avg_raw_score	0.78077
ctime	2026-05-22 22:53:30
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.138807119808647
failure_penalty	-0.0
failure_visits	0
id	10
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	10
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.0
validated_reward	2.34231
validated_visits	3
validated_weight	1.0
validation_score	0.7808
visits	3
parent_id	8
child_ids	[16, 12]

Node 11 (evolve)

Property	Value
uct_value	1.1772
avg_raw_score	0.78077
ctime	2026-05-22 23:06:36
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	11
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.0
validated_reward	1.56154
validated_visits	2
validated_weight	1.0
validation_score	0.7808
visits	2
parent_id	9
child_ids	[18]

Node 12 (evolve)

Property	Value
uct_value	1.4821
ctime	2026-05-22 23:18:46
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	12
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.78077
validated_visits	1
validation_score	0.7808
visits	1
parent_id	10
child_ids	[]

Node 13 (evolve)

Property	Value
uct_value	1.6649
avg_raw_score	0.78077
ctime	2026-05-22 23:30:38
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	13
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	13
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.0
validated_reward	0.78077
validated_visits	1
validated_weight	1.0
validation_score	0.7808
visits	1
parent_id	9
child_ids	[]

Node 14 (evolve)

Property	Value
uct_value	1.6074
ctime	2026-05-22 23:42:42
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	evolve
time_step	14
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.85811
validated_visits	1

validation_score	0.8581
visits	1
parent_id	7
child_ids	[]

Node 15 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-23 00:25:20
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	15
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	debug
time_step	15
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	2
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 33-basic_prompt/node_15/conda_env

Resolved 239 packages in 4.01s

Uninstalled 1 package in 1.80s

Installed 237 packages in 56.87s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.14

+ botocore==1.43.14

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.1

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt=... [truncated, 42548 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon failed to train any models because the Ray backend attempted to create UNIX domain sockets with file paths exceeding the system limit of 107 bytes (OSError: "validate_socket_filename failed: AF_UNIX path length cannot exceed 107 bytes"). This is due to excessively long temporary directory

paths used for Ray's inter-process communication sockets, causing all model training attempts to fail.

SUGGESTED_FIX:

Shorten the base directory paths used for temporary files and model output so that the full socket path (including all subfolders created by Ray) stays well below 107 characters. Move your working directory (including data, output, and conda environment) to a location with a much shorter absolute path (e.g., /tmp/ml or /scratch/ml), and rerun the script to avoid exceeding the UNIX socket path length limit.

Node 16 (evolve)

Property	Value
uct_value	1.4821
ctime	2026-05-23 00:32:18
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
failure_visits	0
id	16
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	16
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.78077
validated_visits	1
validation_score	0.7808
visits	1
parent_id	10
child_ids	[]

Node 17 (debug)

Property	Value
uct_value	1.7684
avg_raw_score	0.84677
ctime	2026-05-23 00:43:58
debug_attempts	1
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.36707452725250295
exploration	1.8927326054508324
failure_penalty	-0.0
failure_visits	0
id	17
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.36707452725250295
num_children	1
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have then
stage	debug

time_step	17
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.36707452725250295
validated_reward	1.69589
validated_visits	2
validated_weight	1.0
validation_score	0.8468
visits	2
parent_id	4
child_ids	[19]

Node 18 (evolve)

Property	Value
uct_value	1.1772
ctime	2026-05-23 01:26:05
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	18
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.78077
validated_visits	1
validation_score	0.7808
visits	1
parent_id	11
child_ids	[]

Node 19 (evolve)

Property	Value
uct_value	1.5574
ctime	2026-05-23 01:38:11
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	<pre>### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them</pre>
stage	evolve
time_step	19
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.84912
validated_visits	1

validation_score	0.8491
visits	1
parent_id	17
child_ids	[]